

Live, remote, and disability-inclusive HCI research and practice — An exploratory workshop with representatives from Swedish Disability Rights Organisations

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Abstract

In this exploratory workshop, HCI researchers and practitioners and representatives from Disability Rights Organisations meet to learn from each other's experiences of remote collaboration and communication in hybrid settings. The workshop will be a "crash course" in how live, remote, and disability-inclusive HCI research and practice can be organised for everyone to participate on equal terms, regardless of functional capacity. The focus is not placed on the technologies used but on interpersonal communication, facilitation and management issues, and hands-on design practices. An emphasis will be put on how to foster solidarity among managers and colleagues and how to take collective responsibility for people who often are being marginalised, such as people with disabilities. Based on the outcome of the workshop, a collection of guidelines will be drafted targeting HCI researchers and practitioners who collaborate with stakeholders in remote and hybrid settings.

CCS Concepts

• **Human-centered computing** → **Accessibility theory, concepts and paradigms**.

Keywords

HCI, disability-inclusive, remote, hybrid, Disability Rights Movement

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1 Workshop aims and objectives

In collaboration with representatives from Disability Rights Organisations, this workshop aims to explore and learn how live, remote and disability-inclusive human-computer-interaction (HCI) research and practice can be organised and facilitated for everyone to participate on equal terms, regardless of functional capacity. The focus is not on the technologies used but interpersonal communication, facilitation and management issues, and hands-on design practices. An emphasis will also be put on how to foster solidarity among managers and colleagues and how to take collective responsibility for people who often are being marginalised, such as people with disabilities. With HCI research and practice, we refer to activities conducted in all phases of a research and design process, from discovering and analysing user needs to identifying design opportunities, ideation and prototyping, implementation and user-testing [16].

The objectives of the workshop are:

- to bring HCI researchers and practitioners together with representatives from Disability Rights Organisations to learn from each others' experiences of remote collaboration and communication in hybrid settings,
- to through practical, hands-on design explorations identify disability-inclusive related challenges that arise when working in hybrid settings,
- to capture and reflect upon challenges and experiences gained from partaking in the workshop explorations conducted in hybrid settings,
- to draft guidelines for live, remote and disability-inclusive HCI research and practice.

The guidelines will build on the outcome of the workshop, previous HCI research [4, 6, 9, 13, 17, 18], and the lived experiences of the Disability Rights Organisation representatives.

2 Background and relevance of the workshop

In a post-pandemic world, working life for many people has become a hybrid experience consisting of a combination of work at the workplace and remote or hybrid work from home or other locations using various remote work technologies [4, 8, 17]. Remote work technologies encompass several layers of software and hardware that enable work tasks to be performed remotely in collaboration with other, geographically distributed, colleagues [11]. Such new forms of communication and collaboration in working life have become a "digital normality" [12]. The rapid shift in work practices caused by the COVID-19 pandemic has led to an increase in digital maturity, both in terms of knowledge and experience of working with remote work technologies and upgraded equipment. This shift, both culturally and organisationally speaking, has strengthened the possibility of involving marginalised stakeholders in research

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who otherwise might not have the opportunity to participate in and contribute to HCI research and practice, including people with disabilities.

One of the cornerstones of the Swedish disability policy [15], the UN Convention on the Rights of Persons with Disabilities [2] and the UN's 2030 Agenda for Sustainable Development [3], is the principle that everyone is of equal value and has equal rights. In an inclusive and equal society, everyone should be offered the opportunity to contribute to and participate in the development of society, including HCI research and practice and how technologies are developed and employed in different contexts [10].

The new opportunities (and challenges) offered for conducting live, remote and disability-inclusive HCI research and practice, create conditions for *hyperlocal* experiences of people with disabilities to be meditated and shared beyond their local community and become part of a *global* experience. We see openings for increased inclusion and participation — however, this can only happen if we learn *how* to create conditions for *all* to partake in such distributed and disability-inclusive HCI research and practice. Also, there might be a risk of overestimating the opportunities offered by remote practices. What are we missing out on in remote research and design settings? How can such settings truly contribute to disability inclusion? When do remote collaboration and communication become counterproductive?

2.1 Learning from the Disability Rights Movement – "Nothing about us without us"

In our current design research project [19], we collaborate with Disability Rights Organisations in Sweden to explore how we can benefit from learning experiences gained from the COVID-19 pandemic and how these experiences can contribute to increased participation and inclusion of people with disabilities in society – also including the field of HCI. The Disability Rights Movement has a long history of working with issues related to people with disabilities' rights to participate in and contribute to society [5, 7]. "Nothing about us without us" [7] is a motto pointing out that no design, policy- or decision-making should be developed without the direct participation of the people affected by a design, or a policy.

Together with representatives from Disability Rights Organisations, we have conducted a series of workshops in hybrid settings and experimented with different forms of engagement and collaboration. What we have learned so far is that remote and hybrid settings hold the potential to involve marginalised groups. However, we have also learned how complicated it is to meet the needs of all participants in, for example, hybrid video meetings where local participants are co-located around one device and remote participants partake from other locations. This goes for people with disabilities as well as those without.

2.2 Remote and disability-inclusive research and practice

We currently see a growing body of research exploring disability-inclusive hybrid meeting experiences. Up to now, much of the efforts have been placed on technology concerns, and how remote work technologies can be designed and modified to accommodate people with special needs but we also see research on in-person practices to further equality [1, 6, 13]. Compared to physical space, collaboration and communication in remote and hybrid settings become complicated due to difficulties in interpreting signals in facial expressions, lack of body language and non-verbal communication, latency/delay in communication etc. [13, 14]. Bodily and spatial interaction is missing since signals connected to the space and the body are not made possible in video meetings where only parts of the body are represented on the screen and not in relation to others. In remote and hybrid meeting settings we are all to some extent made disabled since our abilities to communicate and express ourselves through all our senses are limited.

Other challenges faced in disability-inclusive remote and hybrid settings are related to leadership and organisational skills among managers, meeting organisers and facilitators in, for example, co-design workshops [14]. As our research shows, reaching disability inclusion is also a question of solidarity and collective responsibility among peers [14]. To people with disabilities, it might be more challenging to keep up with the quick pace of remote and hybrid meetings and take the floor, especially in larger groups of people. Remote work technologies, such as conferencing platforms also tend to be quite difficult to navigate, which causes confusion and insecurity among the participants about how to break into and become part of discussions. It is one thing to receive information at a meeting or workshop, but it is another thing to be part of a collaboration, a conversation or ask for permission to speak. These difficulties often result in people staying quiet in the background.

There is a knowledge gap and a lack of managers and colleagues who acknowledge these difficulties in communication and collaboration in hybrid settings, and actively invite and give space to people with disabilities [14]. Instead, people with disabilities more often have to fight for themselves. Being that person can be exhausting and result in people getting tired, quiet, and eventually excluded. Managers and colleagues ought to collectively contribute to creating conditions for people with disabilities to partake, instead of leaving them on their own with a feeling of always being the ones who complain and ask for alternative solutions and "special treatment".

This is however nothing new. People with disabilities have at all times experienced exclusion in society and social and participatory events. They have thus been forced to find solutions to the challenges faced in whatever context, also in remote and hybrid work settings. As experienced in our collaboration with the Disability Rights Organisations and also put forward in previous research [9, 14], we still have far to go to achieve conditions for people with disabilities to participate and contribute on equal terms in hybrid and remote research interventions to reach a greater inclusion in research and knowledge production.

What can the HCI community learn from the Disability Rights Movement? How can the experiences of people with disabilities contribute to improving the conditions for live, remote and disability-inclusive HCI research and practice for *all* (people with and without disabilities)?

3 Workshop activities and format

We invite participants from the HCI community and representatives of Disability Rights Organisations to participate in a half-day hybrid

design workshop with local and remote participants. A video conferencing platform (Zoom) and shared online tools for joint sketching will be used for communication and collaboration. The workshop will be a "crash course" on how live, remote and disability-inclusive activities can be organised, what challenges arise and how they can be handled.

3.1 Workshop structure

The workshop will include an introduction, three collaborative activities, a final presentation and discussions. The participants will be divided into three groups with onsite and online participants, and given a simple design task to work on (yet to be formulated in detail). They will spend 40 minutes working on the task using the tools introduced by the organisers. Upon completing the task, the participants will reflect upon the experiences of working together, and what went well and not so well. A reflection guide is provided to structure the discussions. The participants will thereafter gather in a plenum to discuss their experiences. The session ends with a joint reflection and a draft of concrete guidelines and principles to follow when aiming for disability inclusion in remote HCI research and practice.

As obvious, the list of experiences is not a finalised set of guidelines but a start of a methods kit for inclusive and remote HCI research and practice.

The workshop organisers have extensive experience organising collaborative workshops in online, onsite and hybrid settings. A first draft of the workshop timeline is presented in Table 1, which will be further developed and adjusted to fit the conference site and the registered participants.

3.2 Intended audience

The workshop will bring together HCI researchers and practitioners with representatives from Disability Rights Organisations to share learning experiences and best practices. The workshop will be of special interest to HCI researchers, designers and practitioners who are interested in learning and developing methods and practices for organising and facilitating disability-inclusive HCI research and practices.

We expect at least 20 participants in the workshop, including six people representing Disability Rights organisations who will be recruited by the workshop organisers.

3.3 Call for participation and promotional strategy

A call for participation will be published on the project website: https://dworklife.uni.mau.se/live-remote-and-disability-inclusive-hci-research-and-practice/. When signing up for the workshop, the participants are asked to submit a 300-word motivation for joining the workshop. The motivation can, for example, be about their previous experiences of HCI research and practice in hybrid settings, the opportunities and challenges they faced, or an urge for competency development.

Invitations to the workshop will be sent and shared via design-related mailing lists and online communities, for example, Linkedin, NordiCHI, SIGCHI, and via the Disability Rights Movement and the Disability Rights Organisations we are collaborating with.

4 Expected outcomes

The workshop will result in increased competencies among the workshop participants about how to facilitate and organise disability-inclusive, remote HCI research and practice. Their new learning experiences will benefit not only people with disabilities but *all* people they collaborate with in hybrid settings. As mentioned, in remote and hybrid meeting settings we are all to some extent made disabled since our abilities to communicate and express ourselves through all our senses are limited.

The outcome of the workshop will be documented in a workshop report and published as open access. Based on the outcome, a collection of guidelines will be drafted targeting HCI researchers and practitioners who collaborate with stakeholders in remote and hybrid settings. The draft and experiences gained from partaking in the workshop might also result in a joint publication or a research proposal depending upon the interest among the participants. The workshop participants will also be invited to continue sharing practices and methods after the workshop, to sustain the network and keep working towards disability-inclusive HCI research and practice in hybrid settings.

The ultimate goal is to create a community of practice around remote disability-inclusive HCI research and practice, which can share resources, guidelines, advice etc. on how to achieve conditions for increased participation and inclusion in HCI.

5 Organising committee

Elisabet M. Nilsson, Associate professor in Interaction Design at the School of Arts and Communication (K3), Malmö University, Sweden. Through participatory action research, she explores conditions for human and more-than-human inclusion and participation in design, and how this can be played out in real-world settings. She is currently involved in a research project exploring how disability-inclusive hybrid work environments can be designed and facilitated. Another ongoing research project explores more-than-human-centred design practices and how these could be taught and integrated into future design education curricula. Besides research activities, she teaches at the Interaction Design Master's and Bachelor's Programmes at K3.

Magnus Eriksson, Senior researcher at RISE, Research Institute of Sweden unit of Inclusive System Innovation. He has a background in sociology and (STS) and is leading and participating in a number of interdisciplinary research projects around social and digital trust, citizen empowerment engagement, inclusion and norm-critical perspectives on technology and society. Current research projects explore topics such as resilience and crisis preparedness, systematic perspectives on design research, citizen dialogues and antagonisms in sustainability transitions.

Jörgen Lundälv, Ph.D. Social Policy, Associate professor in Social Work at the Department of Social Work and since 2002 Senior Lecturer at the University of Gothenburg, Sweden. His previous research concerns social policies for disabled people, processes of social exclusion, accessibility, patient involvement and inclusive research. He holds a PhD in social policy from the Department of Social Policy at the University of Helsinki, 1998 (Helsinki, Finland).

Time (3:00 h)	Workshop activities
20 min	Welcome and introduction to the workshop program and a presentation of the participants
40 min	Activity 1: Group work on design task (including a short break)
15 min	Break
40 min	Activity 2: Group work, reflecting on the collaboration
15 min	Break
40 min	Activity 3: Plenum, group presentations about their experiences
10 min	Concluding words, next steps, the network

Table 1: The workshop timeline.

Since 2021, he is the coordinator and platform leader for the research group "Social work, health and ill health" at the University of Gothenburg. He is also an Associate professor in Traffic Medicine at the Department of Surgical and Perioperative Science, Umeå university, Sweden.

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